

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`

D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Answers:

SCENARIO: WEB DEVELOPMENT STANDARDS IMPLEMENTATION

Introduction

A startup is developing a modern web application and faces different issues related to HTML5 elements, HTTP methods, XML syntax, and web communication protocols. By applying web development standards, the team can create a reliable, secure, accessible, and standards-

compliant application.

The correct options are identified and justified below.

1. VIDEO EMBEDDING

Given Options

A.

```
<media src="video.mp4"></media>
```

B.

```
<video src="video.mp4" controls></video>
```

C.

```
<movie src="video.mp4"></movie>
```

D.

```
<vid src="video.mp4"></vid>
```

Correct Answer: B. <video>

```
<video src="video.mp4" controls></video>
```

Justification

The HTML5 <video> element is the standard element for embedding video content directly into a web page.

The src attribute specifies the video file:

```
src="video.mp4"
```

The controls attribute displays built-in video controls such as:

- Play
- Pause
- Volume
- Video progress
- Full-screen option

A more robust implementation is:

```
<video controls width="640">
```

```
<source src="video.mp4" type="video/mp4">
```

Your browser does not support the video element.

```
</video>
```

Why Other Options Are Incorrect?

Option	Reason
<media>	Not a standard HTML5 video element
<movie>	Not a standard HTML element
<vid>	Not a valid HTML5 element
<video>	Standard HTML5 element for video embedding

Conclusion

The correct choice is:

B. <video src="video.mp4" controls></video>

because <video> is the standard HTML5 element for embedding video content.

2. FORM SUBMISSION METHODS

Given Options

- A. GET
- B. POST
- C. PUT
- D. DELETE

Correct Answer: B. POST

For a registration form that sends user data to the server, the appropriate method is:

```
<form action="/register" method="POST">
```

Justification

The POST method sends form data in the HTTP request body rather than adding it to the URL.

Example:

```
POST /register HTTP/1.1
```

```
Content-Type: application/x-www-form-urlencoded
```

The data is sent in the request body:

```
username=Gnaneswari&email=user@example.com
```

Why POST Is Better for Registration?

Registration data may include:

- Name
- Email address
- Password
- Phone number
- Other personal information

POST is preferable because the data does not appear directly in the URL.

Important Security Note

POST alone does not encrypt information. Secure registration must use:

HTTPS + POST

Why Other Options Are Less Suitable?

Method	Typical Usage
GET	Retrieve data; parameters are visible in the URL
POST	Submit new data to the server
PUT	Replace or update an existing resource
DELETE	Delete a resource

Conclusion

The correct choice is:

B. POST

because registration involves sending data to the server for processing and storage.

3. NAVIGATION ELEMENTS

Given Options

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Correct Answer: B. <nav>

The semantic HTML element specifically designed for navigation links is:

```
<nav>
```

```
<a href="/">Home</a>
```

```
<a href="/about">About</a>
```

```
<a href="/services">Services</a>
```

```
<a href="/contact">Contact</a>
```

```
</nav>
```

Justification

The <nav> element represents a section of a page containing navigation links.

It provides meaning to:

- Browsers
- Search engines
- Screen readers
- Assistive technologies
- Developers

Why Semantic HTML Is Important?

Semantic elements describe the purpose of content.

For example:

`<nav>`

clearly indicates that the content is navigation.

A generic:

`<div>`

does not provide this meaning.

Why Other Options Are Not the Best Choice?

Conclusion

The correct choice is:

B. `<nav>`

because it is the standard semantic element for navigation menus.

4. XML SYNTAX

Given Options

Option A

```
<note><to>User</to><from>Admin</from>
```

Option B

```
<note><to>User</to><from>Admin</from></note>
```

Option C

```
<note><to>User</from></note>
```

Option D

```
<note to="User"></note>
```

Correct Answer: B

```
<note>
```

```
<to>User</to>
```

```
<from>Admin</from>
```

```
</note>
```

Justification

XML requires:

1. Every opening tag must have a matching closing tag.
2. Elements must be properly nested.

3. XML is case-sensitive.
4. Attribute values must be enclosed in quotation marks.
5. XML documents must be well-formed.

The correct structure is:

```
<note>
```

```
<to>User</to>
```

```
<from>Admin</from>
```

```
</note>
```

The root element is:

```
<note>
```

and it is properly closed with:

```
</note>
```

Errors in the Other Options

Option A

```
<note><to>User</to><from>Admin</from>
```

Error: The root element `<note>` is not closed.

Correct:

```
</note>
```

Option C

`<note><to>User</from></note>`

Error: The tags are mismatched.

The opening tag is:

`<to>`

but the closing tag is:

`</from>`

The correct structure would be:

`<to>User</to>`

Option D

`<note to="User"></note`

Error: The closing tag is incomplete because it is missing the closing angle bracket:

`>`

Correct:`<note to="User"></note>`

Conclusion

The correct answer is:

B. `<note><to>User</to><from>Admin</from></note>`

because it is properly nested and all elements are correctly closed.

5. WEB COMMUNICATION PROTOCOLS

Given Options

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Correct Answer: B. HTTP

The correct protocol for communication between a web browser and a web server is:

HTTP – HyperText Transfer Protocol

Example:

Browser → HTTP Request → Web Server

Browser ← HTTP Response ← Web Server

Example HTTP Request

GET /index.html HTTP/1.1

Host: example.com

Example HTTP Response

HTTP/1.1 200 OK

Content-Type: text/html

Why HTTP Is Used?

HTTP is designed for:

- Requesting web pages
- Sending form data
- Retrieving resources
- Communicating between browsers and web servers
- Transferring HTML, CSS, JavaScript, images, and other web content

For secure communication, the application should use:

HTTPS

HTTPS is HTTP with encryption provided by TLS.

Why Other Protocols Are Incorrect?

Protocol	Primary Purpose
FTP	File transfer
HTTP	Web communication
SMTP	Sending email
SNMP	Network management and monitoring

Conclusion

The correct choice is:

B. HTTP

because HTTP is the standard protocol used for communication between web clients and web servers.

FINAL ANSWER SUMMARY

Question	Correct Option	Concept
1	B	<video> for video embedding
2	B	POST for registration form submission
3	B	<nav> for semantic navigation
4	B	Properly formed XML structure
5	B	HTTP for web communication

OVERALL CONCLUSION

The startup should follow established web development standards to create a modern and reliable application.

The correct implementation is:

- Use the HTML5 <video> element for promotional videos.
- Use the POST method for submitting registration data.
- Use the semantic <nav> element for navigation menus.
- Ensure XML documents are properly nested and all tags are closed.
- Use HTTP for web communication and HTTPS when secure communication is required.

Applying these standards improves **compatibility, accessibility, security, maintainability, and reliability** of the web application.

Code of Conduct:

I T Ganeswari(192372183) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T Ganeswari

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand